# Data-Driven Modeling and Predictive Analysis of Global Trade Dynamics using Machine Learning Models




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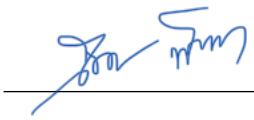
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# Abstract

This thesis studies the issue of data-driven modeling and predictive analysis of global trade dynamics using machine learning models by focusing on international trade between Thailand and GCC countries (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates). The GCC was selected because their countries rely on import food product and have purchasing power whereas, Thailand has the potential to produce the product serve these countries. However, Thailand has not entered into a free trade agreement (FTA) with the GCC countries. Therefore, it is essential to estimate trade potential to understand market opportunities.

The aim of this study was to compare two estimate approaches with the historical gravity model using Poisson Pseudo Maximum Likelihood (PPML) and machine learning models using LASSO, Random Forest, and XGBoost. Then, we used SHAP analysis to find the most important variable, and after that, we chose Moran's I to check the spatial pattern.

Data sources were collected from trade data covering the years 2018 to 2022 from the UN, WDI, CEPII, and Harvard Atlas. Then, we focused on harmonized 6-digit codes for 8 products of halal food and processed food and have a total of 240 observations.

For the results show that machine learning models had significantly outperformed traditional econometric approaches. Especially, Random Forest had the highest accuracy ($R^2 = 0.86$), effectively getting the non-linear trade dynamics more than the traditional PPML model ($R^2 = 0.72$). Additionally, the rule-based tariff logic was constructed to approximate actual trade barriers better than random simulation.

A limitation of this study is the small sample size, tariff data was as created using a rule-based and the time period of the study covered the COVID-19 pandemic, which may not be representative of normal trade. The findings have important implications for policy, especially for the Thailand government to negotiate a free trade agreement (FTA) with GCC countries early and develop different strategies for each country instead of following the same group.

**Keywords:** gravity model, machine learning, PPML, LASSO, international trade, Thailand, GCC.

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# Chapter 1 Introduction

## 1.1 Background

Global trade has changed in recent years due to the effect of geopolitics on logistics [1]. For example, important events such as the trade war between the United States and China have created a barrier to trade by increasing the import tariff rate [2]. Another event is the emergence of new technology platforms that ease trade between businesses to customers or customers to customers [3]. As a result, many countries are trying to change their trade patterns and find new markets to decrease the effects of trade wars.

Thailand is one country that has been affected by these cases because it relies on exports, especially agricultural products, food, and processed food, which are the main products and have potential. These products of Thailand have been widely accepted in terms of quality and safety, which means there is an opportunity to find a new market in the Arab zone for halal food, for which Thailand has a halal standard[4]. Furthermore, Thailand's policy has become known as the "Kitchen of the World" to support businesses in accessing markets and increasing their capacities[5, 6].

The GCC region comprises six countries, such as Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates. All of these countries have high revenue from oil and gas exports but are unable to produce food by themselves due to their desert climate limited water resources, and agricultural land. Consequently, they have to depend on imports from abroad. By 2022, GCC countries imported more than 80% of their food demand [7, 8]. This makes the GCC a high-potential market for Thailand.

In 2022, a significant turning point was the reunion of Thailand-Saudi Arabia relations[9, 10], which marked the end of a breakup lasting more than three decades. This event has opened new opportunities for trade and business cooperation. Furthermore, the GCC countries are implementing food security strategies, such as Saudi Vision 2030 and the UAE Food Security Strategy, to ensure food security [11, 12]. These strategies focus on diversifying import sources and building collabo-

rative alliances with food exporters. Thailand is a country as a major food producer and well-suited for this situation.

However, the trade relationship between Thailand and GCC countries is still not at full potential. In 2022, Thailand has exported food products to GCC countries with high significant growth[13]. Food products are the main share of this trade. Nevertheless, GCC countries still import food from competitors such as Brazil, Australia, and EU countries. The question is why Thailand is not able to capture more of this market share, despite having a competitive advantage.

There are several possible explanations. Firstly, Thailand has not signed a free trade agreement with GCC countries, so tariffs and non-tariff barriers are still present [14]. Secondly, there may be problems with logistics and supply chains. Thirdly, there may be differences in regulations, especially halal certification standards [15]. Fourthly, cultural and business practices can create resistance. This study aims to systematically examine these factors using a quantitative approach.

The focus on halal and processed foods is intentional, as group products are potential producers that meet the continuously high demand from the GCC countries. In addition, Thailand has established capabilities in food production and export markets.

## 1.2 Problem Statement

Although Thailand has a strong capability in halal and processed food, trade with GCC is still not at full potential. Many GCC countries import more food from other competitors, such as Brazil or the EU, than from Thailand. This is surprising because Thailand is closer to the GCC than Brazil and has lower labor costs than EU countries. There must be other factors that traditional models may miss.

Moreover, there is also a technical problem that this study addresses. Trade data is often noisy and has many zeros, which means there is no trade flow between some product-country pairs [16]. This is especially true when looking at data at the HS6 level. Traditional models, such as the Gravity Model with PPML, sometimes cannot handle this complexity [17]. The data may become unstable when the data is noisy. In the basic analysis for this study, PPML often converges after many iterations even when trying different specifications.

Traditional models assume a specific functional relationship between variables such as GDP, distance, and trade flows [18, 19] but real trade in the world is nonlinear and affected by complex factors that have unexpected relationships. For example, the impact of taxes may differ for different types of products, or distance may be less important for higher-value products than consumer goods with large volumes.

Traditional models are not good at capturing these differences.

Machine learning models have the potential to address these challenges [20, 21]. They can handle multi-dimensional data with many zeros and capture non-linear relationships without specifying a functional form in advance. They also have tools like SHAP values to understand which variables drive predictions [22]. However, machine learning models are not yet widely used in bilateral trade analysis, especially in the context of developing countries [23].

Therefore, Thailand needs a more scientific approach to understand these three important questions. First, why do exports vary so much from year to year? Second, can machine learning do better than traditional economic models for trade forecasting? Third, which country in the GCC has the best prospects for Thai exporters when considering the specific characteristics of each country?

## 1.3 Research Objectives

This thesis has five main objectives:

1. The first objective is to build a comprehensive data panel covering the years 2018-2022 by summarizing trade data from UN COMTRADE [24], economic indicators from WDI [25], distance from CEPII [26, 27], and product complexity from Harvard Atlas of Economic Complexity [28]. This is a basic step to prepare and clean the data before analysis for the next step.

2. The second objective is to estimate the gravity model using PPML to identify the most important factors of Thai-GCC trade [16]. Although PPML is not a good model, it can provide valuable information about the data.

3. The third objective is to build machine learning models such as LASSO [29], Random Forest [30], and XGBoost [31]. These models should be more accurate in prediction and better at handling complex data patterns than traditional models.

4. The fourth objective is to analyze variable importance using SHAP values [22] and check spatial patterns by using Moran's I [32, 33]. SHAP analysis separates the prediction into parts supported by each variable to help identify which factors have the most influence on Thai-GCC trade.

5. The fifth objective is to discuss trade policy. The findings should provide deep insights that can be implemented for Thai government officials and business associations working on market expansion.

## 1.4 Research Questions

This research has the following questions:

1. What are the most important factors between economics factors and policy factors are affect to Thai-GCC trade? This question is a basic question to all analysis in this thesis, making it easier to understand which factors are most important to Thai-GCC trade for supporting academic and related policy-making.

2. Can machine learning models outperform traditional PPML models [16]? This is a methodological question with implications that extend beyond this specific study.

3. Which GCC market shows different patterns of trade from what the standard model predicts? This question is important for identifying anomalies that need further investigation.

4. How important are digital readiness and distance in explaining trade patterns [19]? This question is important for understanding factors that have a special relationship with digitalization and platformization going to change in the pattern of trade.

## 1.5 Contributions

This study contributes to both academic knowledge and practical policy design.

Academically, this study bridges the gap between econometrics and data science. Most studies use historical econometric models to estimate the gravity model but do not compare with machine learning models [23]. This study uses both econometric and machine learning models to estimate the gravity model and compare the results by using Thai-GCC trade as a case study [34, 35]. The results are reasonable and have methodological significance for future research.

The study also contributes to understanding South-South trade dynamics. Most studies of the gravity model focus on developed countries and developing countries but not on South-South trade, which means low-income and middle-income economies or between developing economies [17]. The trade between Thailand and the GCC is a unique pattern, different from others, and should be of more interest.

From a policy perspective, this study provides insights for future trade negotiations. We found that the tariff is the most important variable to support FTA negotiations [14]. It was also found that each country in the GCC has a different

pattern of trade, and it is not appropriate to use the same approach for all countries. Therefore, it is essential to have a country-specific approach.

## 1.6 Thesis Organization

This thesis is organized into six chapters.

Chapter 2 presents the background of the study. We issued the foundation by discussing the basic theoretical foundation of the gravity model [36, 37] and machine learning approaches [38]. As well as the economic relationship between Thailand and the GCC and the halal food industry.

Chapter 3 provides a comprehensive Literature Review. We review previous research on gravity models [23, 19] and machine learning approaches in trade analysis [20, 21], and identify gaps in the literature that this study addresses.

Chapter 4 details the methodology. We explain the research design, data collection process and analysis techniques. The data sources include UN COMTRADE [24], WDI [25], CEPII [26], and Harvard Atlas [28]. The estimation methods include PPML [16], LASSO [29], Random Forest [30], and XGBoost [31].

Chapter 5 presents the results and discussions. We analyze the findings in the context of the research objectives and research questions, including descriptive statistics and model comparison results and variable importance analysis.

Chapter 6 concludes the thesis. We summarize the main findings and their implications for policy recommendations, acknowledge the limitations of the study, suggest future research direction, and follow up with policy recommendations. It acknowledges the limitations of the study and suggests future research directions and follows up with policy recommendations.

# Chapter 2 Background

This chapter provides the background for understanding trade between Thailand and the GCC, covering world trade dynamics and the structure of Thailand's exports. It also explains the economic nature of the GCC and provides details from the literature review of international trade and the digital trade report. It is important to understand this chapter before proceeding with quantitative analysis.

## 2.1 Overview of Global Trade Dynamics

Global trade has changed significantly over the past decade and has many new factors and challenges affecting international trade today [1].

For instance, the COVID-19 pandemic has affected the global supply chain. Serious geopolitical issues like trade barriers and trade wars have increased uncertainty in trade that makes the trend of the supply chain moving back to its original country. Digital technology has changed the way businesses operate in various countries, including consumer behavior, such as online shopping. And climate change has affected agricultural production.

Countries now focus on two main goals. First, creating an elastic supply chain. Second, diversifying export markets. These goals are important for Thailand because it depends on exports. When one market closes, we need a new choice.

The World Trade Organisation reported that global trade grew only 2.7 percent, lower than the average of about 5 percent in the past [1]. The slowdown reflects uncertainty, so trade policies have become more politicised. Therefore, the main economies use taxes as a strategic tool.

The measure of a country's trade openness, which represents its connection to the world economy, is this formula [19]:

$$Trade_Openness = \frac{Exports + Imports}{GDP} \tag{2.1}$$

Thailand has a high trade openness of around 120 percent. This means they are well-connected to the world economy [25]. When global trade slows down, they are immediately feel the impact. Table 2.1 shows the comparison with other countries.

Table 2.1: Trade Openness Comparison

| **Country** | **Trade Openness (%)** |
|---|---|
| Thailand | 120 |
| Vietnam | 200 |
| Malaysia | 130 |
| Indonesia | 37 |

Regional trade agreements have become more important [14]. Thailand joined RCEP in 2022, which is the largest free trade agreement in the world, covering 15 countries and 30% of global GDP. However, Thailand does not have a Free Trade Agreement (FTA) with the GCC countries, so this is a gap that needs to be addressed.

## 2.2 Thailand's Position in International Trade

Thailand is well-known as an export-oriented economy. The important thing is that Thailand is known as the Kitchen of the World[39], with the support of government policy, which reflects our strength in food production. Our main export sectors include (1) food and agricultural products, (2) electronic and electric products, (3) parts of vehicles and vehicles, (4) rubber and plastic products, (5) textiles and yarn, (6) leather and accessories, and (7) chemical products.

The government promotes the Thailand 4.0 policy as a strategy to support innovation and automation in the food industry. For food, this means moving from raw materials to processed products. For example, instead of selling only raw chicken meat and rice, Thailand sells ready-to-eat frozen chicken and frozen rice products.

Thai food exports reached about 30 billion USD in 2022, making Thailand one of the leading food exporters in Asia, but there is still room for growth [24, 13]. Table 2.2 shows our main food export categories.

Table 2.2: Thailand's Main Food Export Categories

| **Category** | **Value (Billion USD)** |
|---|---|
| Seafood | 8.5 |
| Rice | 4.2 |
| Processed Food | 6.3 |
| Fruits & Vegetables | 3.8 |
| Sugar | 2.5 |

Thailand has several key strengths in food export. Table 2.3 shows our main food export categories.

Table 2.3: Thailand's Key Strengths in Food Export

| **Strength** | **Description** |
|---|---|
| Quality | High standards and certifications |
| Price | Competitive pricing |
| Variety | Wide range of products |
| Infrastructure | Modern processing facilities |

However, there are also challenges. Labour costs in Thailand are higher than in Vietnam and Indonesia. The volatility of the exchange rate affects competitiveness. Compliance with new standards requires investment that makes Thai exporters continue to innovate to maintain competitiveness.

## 2.3 Thailand's Halal and Processed Food Industry

Halal food is a major advantage for Thailand in GCC markets, as it ensures that Thailand is not mostly Muslim. They have only 5 percent of the Muslim population, but the country has specialists and a high quality of halal food.

The Central Islamic Council of Thailand provides halal certification, which is accepted worldwide and recognised by organisations under the GCC. At this point, the certification process is so rigorous and reliable that it creates opportunities for Thai exporters by opening the market for halal food in the GCC.

Thailand is one of the world's top halal food exporters[40], ranking in the top 10 globally. This is because their have strong infrastructure in food production, and the system of certification is accepted internationally. Furthermore, the price of the product is lower than that of competitors and they have experience in the service market for Muslims in ASEAN. Moreover, Thailand exports a variety of processed food products to the GCC countries and strong to producing these product for example, canned tuna (Thailand is the number one exporter in the world), frozen chicken and processed meat[41, 42, 43], sauce (chili sauce and fish sauce), snacks, instant noodles, drinks, milk products, prepared food and frozen food, biscuits and bakery products.

The halal food sector in Thailand is worth over 6 billion USD annually and provides employment for over a thousand people, and most of the big manufacturing companies have specific skills in halal production. Table 2.4 shows the process of certification.

Table 2.4: Halal Certification Process

| Step | Description |
|---|---|
| 1 | Application submission |
| 2 | Document review |
| 3 | Factory inspection |
| 4 | Product testing |
| 5 | Certification issuance |

Innovation is ongoing in the Thai halal food industry develop new products such as protein alternatives from plants, health food, sustainable packaging solutions, and premium ready-to-eat halal food.

These innovations help Thailand remain competitive in the high-demand GCC market.

## 2.4 Economic Profile of GCC Countries

GCC stands for the Gulf Cooperation Council, which includes six countries: Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the UAE. They have good cooperation between them, but there are some physical differences.

Why are the GCC countries rich? The answer is that they have oil and gas, and exporting these products makes them have a high GDP per capita. For the reasons stated above, some GCC countries have the highest income in the world. However, they have one big problem because they cannot grow enough food.

The reason they cannot grow enough food is because of the weather in the desert, which is too hot (over 45 degrees Celsius), limited water resources, and rainfall that makes their soil is not suitable for most crops. Moreover, they have small agricultural land and labour costs for farms

GCC countries import 80 to 90 per cent of their food, which is a vulnerability in food security [44]. In the case of the COVID-19 pandemic, it shows that these countries are sensitive to a lack of supply chain of food. Table 2.5 shows basic data for each country.

Table 2.5: GCC Countries Economic Profile

| Country | Population (M) | GDP per capita (USD) | Food Import (%) |
|---|---|---|---|
| Saudi Arabia | 35 | 23,000 | 80 |
| UAE | 10 | 43,000 | 90 |
| Qatar | 3 | 61,000 | 90 |
| Kuwait | 4 | 32,000 | 85 |
| Oman | 5 | 18,000 | 80 |
| Bahrain | 1.5 | 25,000 | 85 |

Source: World Bank WDI (2022)

Each GCC country is different, so understanding these differences helps exporters:

**Saudi Arabia** is the largest market with 35 million people, and it is a conservative market, which means the strictest halal requirements exist here. However, it has the most opportunities.

**The UAE** is the center of logistics in the region. Dubai has more imports from other countries, and it is an export hub in the region. As a result, many products prefer to pass through Dubai to go to other GCC countries.

**Qatar** has a high GDP per capita, ranking first in the world. Therefore, the people of Qatar tend to pay a premium price, as quality is considered more important than price.

**Kuwait** has a high demand for Asian food. Thailand is a country that is already popular in this region, and there is a distributive market for expansion.

**Oman** has a policy to position the country as a bridge between the GCC and Southeast Asia. Therefore, they have a lower income per capita than other GCC countries, but the country's middle class is growing. As a result, they have purchasing power.

**Bahrain** is a small market in the GCC countries, but it is easy to enter and is good for testing new products before expanding to larger markets.

## 2.5 Food Consumption Trends in GCC

GCC consumers have specific preferences, so Thai exporters can succeed if they understand these preferences such as Easy-to-prepare food, High-quality products and packaging that are beautiful, Halal certification from recognised organisations, Brands from other countries that they trust, Long shelf life for storage, health and food for health, and products are cost-effective.

These preferences match Thailand's capabilities because it is good at producing processed food and has halal certification. In addition, these products are of high quality and cost-effective.

GCC consumers are also changing behavior to purchasing products. For example, young people like convenience, online shopping is growing fast, and social media has an impact on food choices, which makes these trends benefit exporters with innovation.

GCC countries invest heavily in food security programs [45]. For example, Saudi Vision 2030 has goals for food security, and the UAE has a National Food Security Strategy[46, 47]. These goals include diversifying food imports to reduce risk, building strategic food reserves for sustainable development, developing food production technology, and being a partner with trusted food exporters.

Thailand is suitable for these strategies because it is a reliable supplier with high-quality products.

## 2.6 Diplomatic Developments in 2022

A major event happened in January 2022, when Thailand and Saudi Arabia restored diplomatic relations. This happened after a 30-year break in diplomatic relations, which is considered to be one of the most important diplomatic developments for Thai trade.

Why was the relationship broken? There was a diplomatic incident in 1989 involving the theft of the Saudi royal family's jewels from the Saudi royal family. After that, Saudi Arabia reduced its diplomatic relations with Thailand. However, trade was not banned, but it was very difficult to trade between the two countries.

Before 2022, Thai businesses experienced difficulties because it was difficult to request a visa for business purposes and there were no direct flights between countries. In addition, the government has limited support because Saudi Arabia was not a partner with Thailand, and there were complex bank transactions.

What happened in 2022? It has many events happening this year. The important thing is Saudi Arabia's Crown Prince Mohammed bin Salman visited Thailand, and the Prime Minister of Thailand visited Saudi Arabia. Then, the two sides signed a trade and investment agreement. and have collaborated in business, such as when direct flights between Bangkok and Riyadh were established, business visas became easier, and the two sides established a business council.

This is a game-changer for Thai exporters after the Saudi Arabian market fully opened again. Saudi Arabia is the largest economy in the GCC and has 35 million consumers, which is expected to grow significantly in the years to come.

The restoration also improves Thailand's position with other GCC countries because it sends a signal that Thailand is serious about this region. Other GCC countries may be more willing to do business with Thailand.

## 2.7 Trade Barriers: Tariffs and non-tariff measures (NTMs)

Selling to the GCC is not easy because there are two types of barriers. The first is tariffs, and the second is Non-Tariff Measures (NTMs), both of which increase costs for exporters [17].

GCC applies a common external tariff by using the standard rate, which is approximately 5 percent for most food products[48]. This is nearly the lowest compared to the external region, but Thailand does not have a Free Trade Agreement (FTA) with the GCC. So, they must pay at the Most Favored Nation (MFN) rate, while countries that have an FTA can pay less than the MFN rate or equal zero.

Table 2.6 shows a comparison of Thailand with its competitor.

Table 2.6: Tariff Comparison: Thailand vs Competitors

| **Country** | **FTA with GCC** | **Tariff Rate (%)** |
|---|---|---|
| Thailand | No | 5 (MFN) |
| India | No | 5 (MFN) |
| Australia | Yes | 0-2 |
| New Zealand | Yes | 0-2 |

Normally, NTMs are more difficult than tariffs because they have hidden costs [15]. Table 2.7 shows the types of NTMs in GCC countries.

Table 2.7: Types of Non-Tariff Measures (NTMs) in GCC countries

| **NTM Type** | **Description** |
|---|---|
| SPS | Sanitary and Phytosanitary measures |
| TBT | Technical Barriers to Trade |
| Labelling | Arabic labelling requirements |
| Certification | Halal certification requirements |
| Documentation | Import permits and licenses |

NTMs create extra costs for exporters. They must follow the steps: (1) Pay for halal certification and inspection, (2) create Arabic labels, (3) request health

certificates for each shipment, (4) sometimes pay for product testing, and (5) manage documentation requirements.

Trade cost can be written as [18]:

$$Total_Trade_Cost = Tariff + NTMs + Logistics + Documentation \quad (2.2)$$

This studies suggest that decreasing the cost of NTM is important for trade because their cost equals 10-30 percent[17], while the cost of tariffs equals 5 percent. So, reducing NTM costs is important for trade.

## 2.8 Digital Readiness and Trade Facilitation

Technology is changing international trade by using digital tools, which can reduce trade costs significantly [3]. Examples include the Single Window system, the paperless trade system, the AI inspection system, the digital logistics tracking system, the food product certification system, and the blockchain system for supply chain transparency.

GCC countries are moving towards digital government by investing heavily in technology. For example, the UAE has a high-level Single Window system where all trade documents can be sent online, and Saudi Arabia is modernising its customs as part of Vision 2030 and can reduce the time of the process.

Thailand is also improving its digital infrastructure by having a service as a National Single Window system where all certificate of origin documents can be sent online, and most services are moving more online, but it needs to be edited to be more user-friendly.

When both countries have good digital systems, trade becomes easier, the process is quick, the cost is low, and errors in the process are reduced. So, we can estimate that reducing the distance can be done effectively [19]:

$$Effective_Distance = Physical_Distance \times e^{-\lambda \cdot Digital_Index} \quad (2.3)$$

The parameter lambda measures how digital readiness reduces trade cost. A country with higher digital readiness can trade more easily, even with physical distance.

Digital readiness is important for Thailand-GCC trade, as the physical distance is about 5,300 km. A good digital system can reduce the cost of this distance to benefit both countries from investment in basic digital trade infrastructure.

## 2.9 Conclusion

This chapter explained the background of Thailand-GCC trade. The main issue is GCC countries are rich from oil but do not grow food, which means they must import 80-90 percent of their food needs, thereby creating a large market opportunity.

- Thailand has strong capabilities in halal and processed food. They are a leading exporter of halal food in the world, which aligns with the demand of the GCC.
- The recovery of trade with Saudi Arabia in 2022 opens up new opportunities because the largest market in the GCC can be accessed.
- The trade barriers are high, and the cost of NTMs is more expensive than tariffs, so reducing these barriers will increase trade.
- Digital readiness can reduce trade costs for both parties, and investment in basic digital trade infrastructure benefits everyone.

This background prepares us for the next chapter, and we will review the literature on gravity models and machine learning in trade analysis. After that, we will see how other researchers study similar trade relationships.

# Chapter 3 Literature Review

This chapter reviews the key theories and previous studies and covers seven main topics, namely the classical gravity model of trade, PPML estimation, product complexity, machine learning, SHAP analysis, trade barriers, and digital trade facilitation. The goal is to demonstrate that the knowledge available supports the hybrid approach used in this study.

## 3.1 Classical Gravity Model of Trade

The gravity model is one of the most famous frameworks in international trade because this concept originates from Newton's law of gravity, which explains that the force between two objects depends on their mass and distance. Tinbergen (1962) [36] was the first to apply this concept to trade flows.

The classical gravity model is

$$Trade_{ij} = G\frac{GDP_i^{\alpha} \cdot GDP_j^{\beta}}{Distance_{ij}^{\gamma}} \tag{3.1}$$

This equation shows that trade between countries i and j depends on three factors:

- The economic scale (GDP) of the two countries
- The distance between the two countries
- G is a constant that captures other factors

Larger economies trade more with each other, and if the distance is high, trade is low. So, this idea can explain most patterns of trade in the gravity model.

The log-linear version normally used in the gravity model is

$$\ln Trade_{ij} = \beta_0 + \beta_1 \ln GDP_i + \beta_2 \ln GDP_j - \beta_3 \ln Distance_{ij} + \varepsilon_{ij} \tag{3.2}$$

The researcher estimated this equation by Ordinary Least Squares (OLS), but it has a major problem with heteroskedasticity, which means non-constant error

variance (heteroskedasticity) in OLS leads to biased results.

Table 3.1 shows the development of the gravity model.

Table 3.1: Development of the Gravity Model

| Year | Author | Contribution |
|---|---|---|
| 1962 | Tinbergen | First application to trade |
| 1979 | Anderson | Theoretical foundation |
| 2003 | Anderson & van Wincoop | Multilateral resistance |
| 2006 | Santos Silva & Tenreyro | PPML estimation |
| 2016 | Yotov et al. | Structural gravity guide |

Anderson and Van Wincoop (2003) [18] made an important contribution. They introduced these multilateral resistance terms, which can capture trade costs with all partners affecting bilateral trade. For example, a country does not just consider distance to one partner because it considers distance to all partners.

## 3.2 Poisson Pseudo-Maximum Likelihood (PPML) and Structural Gravity

### 3.2.1 Why PPML is Needed

Santos Silva and Tenreyro (2006) [16] published a very influential paper. The title is "The Log of Gravity." They show that log-linear OLS has serious problems. The main issues are that zero trade values cannot be included (the log of zero is undefined), heteroskedasticity causes bias, and the results depend on how zeros are handled.

They recommend using Poisson Pseudo-Maximum Likelihood (PPML). This estimator has several advantages, such as it can handle zero trade values naturally, is resistant to heteroskedasticity, allows for fixed effects easily, and has better specification testing than OLS.

PPML estimates the model in levels rather than logs:

$$E(Trade_{ij}) = \exp(X_{ij}\beta) \tag{3.3}$$

The WTO-UNCTAD Advanced Guide recommends PPML as the standard method because it can avoid many statistical problems [17]. So, many recent studies like to use PPML for gravity estimation.

### 3.2.2 Structural Gravity

Yotov et al. (2016) [17] published the WTO-UNCTAD Advanced Guide for Trade Policy Analysis. This guide introduces structural gravity with a fixed effects model to control for multilateral resistance.

The general equation is:

$$Trade_{ijt} = \exp(X_{ijt}\beta + \pi_{ij} + \chi_{it} + \psi_{jt}) \tag{3.4}$$

where:

- $\pi_{ij}$ is pair fixed effect
- $\chi_{it}$ is exporter-year fixed effect
- $\psi_{jt}$ is importer-year fixed effect

Table 3.2 summarises the comparison.

Table 3.2: Comparison of OLS and PPML

| **Feature** | **OLS** | **PPML** |
|---|---|---|
| Zero trade values | Cannot handle | Can handle |
| Heteroskedasticity | Biased | Robust |
| Fixed effects | Possible | Easy |
| Interpretation | Elasticity | Semi-elasticity |

## 3.3 Product Complexity and Relatedness

### 3.3.1 Economic Complexity Theory

Hidalgo and Hausmann (2009) [49] developed the economic complexity framework. They argue that products differ in their knowledge requirements, with some products needing many capabilities to produce and other products needing few capabilities.

The Product Complexity Index (PCI) measures, including first, high PCI products, which are products produced by a few countries, require diverse capabilities, and are related to higher incomes. Second, low PCI products are products produced by many countries, require basic capabilities and are found in developing countries.

Many countries are producing complex products for growth, but they cannot skip directly to very complex products because they must build capabilities step by step.

### 3.3.2 Relatedness and Export Potential

Tiits et al. (2024) [34] combine relatedness with gravity models. They show that countries are more likely to export products that are "nearby" in the product space, which means that the products require similar capabilities of production.

This is relevant for Thailand-GCC trade. Thailand's halal and processed food exports belong to related products such as canned fish (HS 1604), processed meat (HS 0207), sauces and condiments (HS 2103), and ready-to-eat food (HS 1905).

Thailand can leverage the existing capabilities of these products, which share capabilities in food processing and halal certification, to expand in GCC markets by using an export strategy with (1) What—products to target, (2) When—time to enter the market, (3) Where—market to prioritize, (4) Who—who should lead, and (5) Worth—expected export value.

## 3.4 Tariffs and Non-Tariff Measures

### 3.4.1 NTMs in Food Trade

Garcés and Vogt (2024) [15] introduce that food trade faces heavy regulation and analyse the form of non-tariff measures (NTMs). They found that agriculture and food are the most regulated sectors. NTMs include sanitary and phytosanitary measures (SPS), technical barriers to trade (TBT), labeling requirements, certification requirements, and inspection procedures.

SPS measures cover 16% of NTMs.

Table 3.3 shows the main types of NTMs.

Table 3.3: Main Types of Non-Tariff Measures

| **NTM Type** | **Coverage** |
|---|---|
| SPS Measures | 16% |
| TBT Measures | 25% |
| Labelling | 12% |
| Certification | 18% |
| Others | 29% |

### 3.4.2 Trade Cost Estimation

Arita et al. (2015) [50] estimate the ad valorem equivalent (AVE) of NTMs for beef products. The AVE of SPS restrictions is 23-24%, and for pork, the AVE is 81%. The result is that these are higher than actual tariffs.

Total trade cost can be expressed as:

$$TotalCost = Tariff + NTMs(AVE) + Documentation + Logistics \tag{3.5}$$

For GCC markets, the main NTM costs are (1) Halal certification required for all meat products, (2) Arabic labels required for all consumer products, (3) storage age requirements for food products, and (4) health certificates required by the government.

### 3.4.3 Regulatory Heterogeneity

Dong et al. (2025) [51] study regulatory heterogeneity in agricultural trade. They find that SPS harmonization under regional trade agreements reduces trade costs, including deep harmonization, which creates trade. While shallow harmonization has little effect.

Countries with similar regulatory systems have more trade between them, but Thailand and GCC countries have different regulatory frameworks that create additional trade costs. However, halal certification provides some good common ground in the future.

## 3.5 Machine Learning in Trade Analysis

### 3.5.1 Why Machine Learning

Traditional gravity models assume linear relationships to explain trade patterns, which can be nonlinear. In this way, machine learning can capture these complex patterns [21]. Recent studies show that ML improves prediction accuracy.

This case study in 2025 compares traditional and ML methods for gravity models [20], with the key findings being:

- XGBoost achieves an R-squared of 0.87 with a log transformation
- Random Forest achieves R-squared 0.86
- OLS achieves R-squared 0.70
- PPML achieves R-squared 0.17-0.20

The result is that ML models outperform traditional methods in prediction accuracy. However, PPML can handle zero trade values better, whereas the choice depends on the research objective.

Table 3.4 compares the methods.

Table 3.4: Comparison of Estimation Methods

| **Method** | **R-squared** | **Zero Handling** | **Interpretability** |
|---|---|---|---|
| OLS | 0.70 | Poor | High |
| PPML | 0.17-0.20 | Excellent | High |
| Random Forest | 0.86 | Good | Medium |
| XGBoost | 0.87 | Good | Low |

### 3.5.2 XGBoost Model

Chen and Guestrin (2016) [31] explain that XGBoost is widely used in trade research, which stands for Extreme Gradient Boosting. The model builds many decision trees, with each tree correcting the errors of previous trees.

The basic model is:

$$\hat{y} = \sum_{k=1}^{K} f_k(x) \tag{3.6}$$

The advantages of XGBoost include handling missing data automatically, capturing nonlinear relationships, providing high accuracy, and working with high-dimensional data, including regularization to prevent overfitting.

### 3.5.3 LASSO and Random Forest

LASSO (Least Absolute Shrinkage and Selection Operator) is useful for variable selection, which shrinks some coefficients to zero to help identify the most important variables [29].

Random Forest builds many decision trees using random subsets of data; thereafter, it averages the predictions to reduce overfitting [30]. So, Random Forest is robust and easy to use.

Both methods provide different perspectives on variable importance and are useful complements to XGBoost.

## 3.6 SHAP and Explainable AI

### 3.6.1 The Interpretability Problem

Machine learning models can be hard to interpret because a neural network has thousands of parameters, which makes it difficult to understand why it makes certain

predictions. We call this problem the "black box".

For policy applications, being able to understand things is important. Trade officials need to know why a model says certain things will happen. They can't make decisions based on predictions that aren't clear.

### 3.6.2 SHAP Values

Shapley Additive exPlanations (SHAP) is a solution based on shapley value from game theory and shows how each variable can predict [22].

The formula is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \tag{3.7}$$

SHAP values have several advantages, such as the sum of SHAP values being the prediction independent of the model, showing the direction (positive or negative), and showing the size of the direction of the contribution of each feature.

The WTO's 2024 AI report presents how to use AI for policy with SHAP analysis, which can help transparency for trade policy decisions [3].

## 3.7 Spatial Analysis in Trade

### 3.7.1 Why Spatial Analysis

The spatial autocorrelation is that trade patterns often show geographic clustering, which means that countries that are close together may have similar trade patterns [33].

Moran's I measures spatial autocorrelation [32]:

$$I = \frac{n}{\sum_i \sum_j W_{ij}} \frac{\sum_i \sum_j W_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \tag{3.8}$$

### 3.7.2 Application to GCC

GCC countries are a small regional group that has only 6 countries in the union. However, they have important geography and are strong to corroborate in terms of similar logistics routes, common customs procedures (GCC Customs Union), similar consumer preferences, and the same food safety standards.

The trade potential with a single GCC nation may be forecast with other nations in the region. That means spatial clustering facilitates the identification of regional patterns.

## 3.8 AI and Digital Trade Facilitation

### 3.8.1 AI in Customs

The WTO and WCO reports highlight several applications. They inform that AI is transforming customs operations [3]. For example, risk management and profiling, fraud detection and compliance, tariff classification, document scanning and verification, and post-clearance audit.

Dubai Customs uses AI-powered applications named The iDeclare app allows travellers to pre-declare items, and users can upload photos. Then AI selects the correct HS code. This speeds up customs processing.

Table 3.5 shows AI applications in customs.

Table 3.5: AI Applications in Customs

| **Application** | **Benefit** |
|---|---|
| Risk profiling | Faster clearance |
| Fraud detection | Reduced revenue loss |
| HS classification | Accurate tariffs |
| Document verification | Less manual work |

### 3.8.2 Digital Readiness

If countries have a complete digital infrastructure, they will have competitiveness of trade cost and make it easier to trade. For example, infrastructure and systems, Electronic Customs (e-Customs) system, paperless trade documents, Single window system for trade, e-certification for products, and internet connectivity.

The GCC countries are grouped by investing in digital infrastructure and completely having advanced systems, such as Saudi Arabia and the UAE. Whereas Thailand is also improving the National Single Window system for use between ASEAN countries.

Digital readiness can reduce the effective distance [19]:

$$Effective_Distance = Physical_Distance \times e^{-\lambda \cdot Digital_Index} \tag{3.9}$$

## 3.9 Research Gaps

The literature review gaps:

(1) **Geospatial Focus**: Most studies examine EU-US or China-ASEAN trade, and there are few studies that focus specifically on Thailand-GCC food trade.

(2) **Method Integration**: Most studies use only one or two methods. However, this study combines PPML + ML + SHAP + spatial analysis.

(3) **Product Level**: Most studies use aggregate trade data, whereas most do not analyse HS6-level trade in halal and processed food.

(4) **Digital Variables**: Digital readiness as a trade determinant is under-researched in gravity models.

(5) **Impact of the 2022**: No study examines the impact of Thailand-Saudi Arabia diplomatic restoration on trade.

This thesis addresses these gaps with focusing on the Thai-GCC halal food trade, Combining PPML with XGBoost and SHAP, Analyzing HS6-level trade in halal and processed food from 2018 to 2022, combining digital variables,and Examining pre- and post-2022 patterns.

Table 3.6: Research Gaps and Contributions

| **Aspect** | **Gap in Literature** | **This Study** |
|---|---|---|
| Geospatial Focus | Few studies on Thailand-GCC food trade | Thai-GCC halal food trade |
| Method Integration | Most use only 1-2 methods | PPML + XGBoost + SHAP + spatial analysis |
| Product Level | Aggregate trade data used | HS6-level halal/processed food (2018-2022) |
| Digital Variables | Under-researched in gravity models | Incorporates digital readiness variables |
| 2022 Diplomatic Impact | No study on Thailand-Saudi restoration | Pre- and post-2022 pattern analysis |

## 3.10 Literature Summary Tables

This section provides summary tables 3.7, 3.8, and 3.9 of the key literature review because these tables help organize the main contributions from different research areas.

Table 3.7: Gravity Model Literature Summary

| **Author(s)** | **Method** | **Key Finding** |
|---|---|---|
| Tinbergen (1962) [36] | OLS | First application of gravity to trade |
| Anderson (1979) [37] | Theory | Provided theoretical foundation for gravity |
| Anderson and Van Wincoop (2003) [18] | Theory + Estimation | Multilateral resistance matters for trade flows |
| Santos Silva and Tenreyro (2006) [16] | PPML | PPML solves heteroskedasticity and zero trade problems |
| Yotov et al. (2016) [17] | Structural PPML | Comprehensive guide for policy analysis |
| Leitão (2024) [23] | Gravity + ML | ML improves prediction over traditional methods |

Table 3.8: Machine Learning in Trade Literature

| **Author(s)** | **Method** | **Key Finding** |
|---|---|---|
| Breinlich et al. (2021) [21] | ML for FTAs | ML captures heterogeneous treatment effects |
| Tiits et al. (2024) [34] | Random Forest | Combines complexity with gravity for prediction |
| Chaffa et al. (2025) [20] | XGBoost vs PPML | XGBoost achieves 0.87 R-squared vs 0.17 for PPML |
| Chen and Guestrin (2016) [31] | XGBoost | Introduced efficient gradient boosting |
| WTO AI Lab (2024) [3] | Various ML | ML useful for trade forecasting |

Table 3.9: NTM and Trade Barriers Literature

| **Author(s)** | **Focus** | **Key Finding** |
|---|---|---|
| Garcés and Vogt (2024) [15] | NTM patterns | Agri-food most regulated sector |
| Arita et al. (2015) [50] | AVE estimation | NTM costs exceed tariff costs for food |
| Dong et al. (2025) [51] | SPS harmonization | Deep harmonization creates trade |
| UNCTAD (2019) [52] | NTM classification | Comprehensive MAST classification |
| WTO (2024) [1] | Trade facilitation | Digital tools reduce NTM costs |

## 3.11 Conceptual Framework

Based on the literature review, a conceptual framework emerges to guide the methodology. The framework has three layers.

### 3.11.1 First Layer: Traditional Gravity

The first layer uses the structural gravity model with PPML estimation [16, 17]. This provides coefficient estimates for key variables (GDP, distance, and tariffs), proper handling of zero trade values, fixed effects to control for multilateral resistance, and a comparison baseline for ML models.

### 3.11.2 Second Layer: Machine Learning

The second layer applied to use machine learning [20, 21]. This captures nonlinear relationships between variables, complex interactions that PPML misses, better prediction accuracy for trade values, and variable importance ranking through SHAP.

Three ML models are used with (1) LASSO for variable selection [29], (2) Random Forest for robust prediction [30] and (3) XGBoost for best accuracy [31].

### 3.11.3 Third Layer: Interpretation and Policy

The third layer focuses on the interpretation of SHAP values to explain model prediction and spatial analysis to identify region layer patterns together. They

provide a clear understanding of what drives trade, identification of trade barriers, policy recommendations based on evidence, and untapped trade potential estimates.

This three-layer framework has process layers built on each other and traditional methods providing the foundation. Then ML adds prediction power and helps with interpretation to ensure policy relevance.

## 3.12 Implications for This Study

The literature review has several implications for this thesis.

**First**, PPML should be the baseline estimation method [16]. Even though it has a lower R-squared than ML, it handles zeros properly because Thailand-GCC trade data likely has zero trade values for some product-country pairs.

**Second**, XGBoost should be the primary ML method [31]. It consistently outperforms other methods; if we combine it with SHAP, it provides both accuracy and interpretability.

**Third**, Thai halal and processed foods have moderate complexity, requiring capabilities in food safety and halal certification, which are transferable across products [49].

**Fourth**, NTMs are likely the main trade barrier because GCC tariffs are low at 5%, whereas NTM costs for food can be 10–30% due to Halal certification and labelling requirements, which add costs [15].

**Fifth**, digital readiness is an emerging factor, and both Thailand and the GCC invest in digital trade systems. Including digital variables may reveal new insights [3].

**Sixth**, the 2022 diplomatic restoration creates a natural experiment, and we can compare trade patterns before and after 2022. So, the findings may show a structural break in Thailand-Saudi Arabia trade.

## 3.13 Conclusion

This chapter reviewed seven areas of literature. Namely, the gravity model provides the theoretical foundation [36, 37], PPML is the recommended estimation method [16], product complexity helps identify export opportunities [49], NTMs create significant trade costs for food products [15], machine learning improves prediction accuracy [20], SHAP provides interpretability [22], and digital readiness is an emerging trade determinant [3].

The literature supports a mixed-method approach and combining traditional and ML methods, including capturing both linear and nonlinear patterns. We found that

SHAP ensures interpretability for policy is most important. This framework guides the methodology in the next chapter.

# Chapter 4 Methodology

This chapter describes the data sources and methods used in this thesis, which covers the data collection process, variable construction, and estimation techniques. The study uses a mixed-method approach by combining PPML gravity estimation with machine learning models. Finally, this study aims to analyse Thailand-GCC trade in halal and processed food products.

## 4.1 Research Design

This study uses a quantitative research design and has four main steps. Figure 4.1 shows the overall research methodology framework four main phases: (1) data collection, (2) data integration and preprocessing, (3) model estimation and training, and (4) analysis and validation.

**Step 1** is collecting four secondary data. These are the UN COMTRADE database for trade flows of eight HS6 products from 2018 to 2022, World Bank World Development Indicators (WDI) for economic variables including GDP, population, and internet penetration, CEPII GeoDist database for bilateral distance data, and Harvard Atlas of Economic Complexity for Product Complexity Index.

**Step 2** is combining data from all sources into a single dataset with 240 observations and 25 variables. Then data preprocessing is done through variable transformation to logarithmic form, legal system coding, and train-test split. Data from 2018-2020 is used for training and data from 2021-2022 is used for testing.

**Step 3** is estimation that uses two types of models. Firstly, the PPML gravity model [16], and secondly, machine learning models, including XGBoost [31], Random Forest [30], and LASSO [29].

**Step 4** is model analysis and validation performance by comparison using $R^2$, RMSE, and MAE metrics. Then, analysis with SHAP values [22] for variable importance interpretation, spatial analysis [33], model diagnostic tests including VIF tests, and residual analysis. There is also an improved process between validation and model training to get the best model performance.

After that, the result can estimate trade flow predictions and policy recommen-

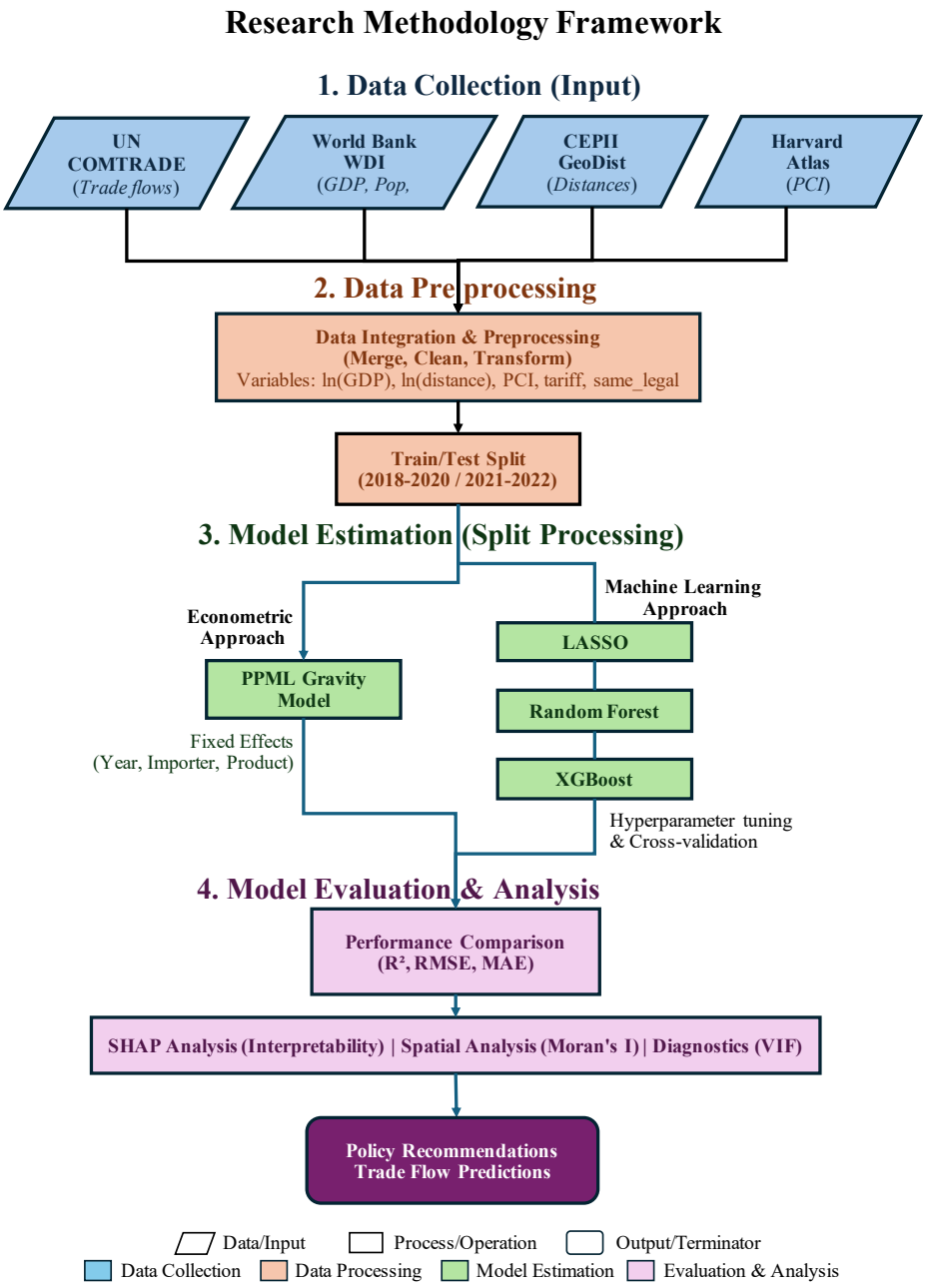



Figure 4.1: Research Methodology Framework

dations that can be used in practice.

The dataset is a balanced panel covering the period from 2018 to 2022. It includes bilateral export data from Thailand to GCC countries. Data comes from four main authoritative sources.

## 4.2 Data Sources

The dataset is a panel covering 2018 to 2022 by merging bilateral export data from Thailand to GCC countries, which form four main sources.

### 4.2.1 UN COMTRADE

UN COMTRADE is the main source for trade data [24]. It is the most authoritative database for international trade in goods by The database covers trade between 250 countries and territories, and it has products classified by 6-digit HS codes.

This study collects export values for 8 selected HS6 products to show Thailand's exports to each GCC country in each year. Trade values are in US dollars.

### 4.2.2 World Bank WDI

World Development Indicators (WDI) provides economic data [25], which has three variables to collect GDP in US dollars, population, and Internet users as a percentage of the population.

Data is collected for Thailand and all six GCC countries in the time period matching the trade data from 2018 to 2022.

### 4.2.3 CEPII Gravity Dataset

CEPII provides geographic and cultural variables [26, 27], which are widely used for gravity model estimation. Variables include bilateral distance between capitals, common language indicator, and legal system origin.

The distance data uses a population-weighted method to ensure consistency across countries.

### 4.2.4 Other Sources

Additional data comes from Harvard Atlas of Economic Complexity for Product Complexity Index [28], WTO for tariff rates [1], and The Thai halal certification status.

Table 4.1 summarises the data sources.

Table 4.1: Summary of Data Sources

| **Source** | **Variables** | **Coverage** |
|---|---|---|
| UN COMTRADE | Trade values | 2018-2022 |
| World Bank WDI | GDP, Population, Internet | 2018-2022 |
| CEPII | Distance, Language, Legal | Cross-sectional |
| Harvard Atlas | Product Complexity Index | 2018-2022 |

## 4.3 Sample Selection

### 4.3.1 Countries

The exporter is Thailand, and the importers are six GCC countries, such as Saudi Arabia (SAU), United Arab Emirates (UAE), Qatar (QAT), Kuwait (KWT), Oman (OMN), and Bahrain (BHR).

These countries form the Gulf Cooperation Council (GCC) and have similar characteristics. All countries have high incomes and import most of their food.

### 4.3.2 Products

HS6 of 8 products, as we selected, represents Thailand's major halal and processed food export by following three criteria.

**First**, the product must have halal potential, which means it can be certified as halal for Muslim consumers.

**Second**, Thailand must have export capacity because these products should be part of Thailand's existing food industry.

**Third**, there must be GCC demand due to the product matching GCC consumer preferences.

Table 4.2 shows the selected products.

Table 4.2: Selected HS6 Products

| **HS6 Code** | **Product Description** |
|---|---|
| 020714 | Frozen cuts of chicken |
| 160100 | Sausages and similar products |
| 160232 | Prepared chicken meat |
| 040210 | Milk powder |
| 190531 | Sweet biscuits |
| 200599 | Vegetables preserved |
| 210390 | Sauces and preparations |
| 210690 | Food preparations |

### 4.3.3 Time Period

The study covers 2018 to 2022, and these five years are chosen for several reasons.

This is because it includes the period before and after COVID-19 and also captures the 2022 diplomatic restoration between Thailand and Saudi Arabia. Otherwise, the UN Comtrade data has good data availability in this period, too.

### 4.3.4 Dataset Structure

The final dataset is a balanced panel with the structure of 6 countries of import in 8 products, covering 5 years, so the total observations = $6 \times 8 \times 5 = 240$.

## 4.4 Variable Construction

### 4.4.1 Dependent Variable

The dependent variable is trade value, which measures Thailand's export value to each GCC country for each product in each year by the unit in US dollars.

For the gravity model, the natural log is used. This is written as:

$$ln_trade_{ijpt} = \ln(1 + TradeValue_{ijpt}) \tag{4.1}$$

The value 1 is added to handle zero trade observations [16].

### 4.4.2 Independent Variables

The independent variables are grouped into four categories as shown in Table 4.3.

Table 4.3: Variable Definitions by Category

| Category | Variable | Definition | Expected Sign |
|---|---|---|---|
| Dependent | ln_trade | Log of export value (USD) | DV |
| Economic Variables | ln_gdp_ex | Log of Thailand's GDP. Represents supply capacity. | + |
| | ln_gdp_im | Log of importer's GDP. Represents demand capacity. | + |
| | ln_pop_im | Log of importer's population. Represents market size. | + |
| Geographic and Cultural Variables | log_distance | Log of bilateral distance in kilometers. Expected to have a negative effect. | − |
| | same_legal | Dummy variable taking value 1 if countries share the same legal system origin. | + |
| Policy Variables | tariff | Applied tariff rate for the specific product. Expected to reduce trade. | − |
| Complexity and Digital Variables | pci | Product Complexity Index. Measures the sophistication of the product. | +/− |
| | internet_im | Internet penetration rate in the importing country. Represents digital trade readiness. | + |

## 4.5 Estimation Methods

### 4.5.1 PPML Gravity Model

The first method is Poisson Pseudo-Maximum Likelihood (PPML). This is the recommended estimator for gravity models, based on Santos Silva and Tenreyro (2006) [16], which shows that PPML handles two common problems that can include zero trade values and is robust to heteroskedasticity.

The model specification is:

$$\begin{aligned} Trade_{ijpt} = \exp\big(&\beta_0 + \beta_1 \ln GDP_{it} + \beta_2 \ln GDP_{jt} \\ &- \beta_3 \ln Distance_{ij} - \beta_4 Tariff_{ijpt} \\ &+ \beta_5 PCI_p + FE\big) \end{aligned} \tag{4.2}$$

Where FE represents fixed effects [17]. The model uses three types of fixed effects:

- Year fixed effects to control for time trends
- The importer fixed effects to control for country characteristics.
- Product fixed effects to control for product characteristics

The model is estimated through maximum likelihood. Standard errors are grouped by pairs of countries.

### 4.5.2 Machine Learning Models

Three machine learning models are used [20, 21]. They capture nonlinear patterns that PPML may miss in the model.

**1. XGBoost stands for Extreme Gradient Boosting [31]**, which creates more decision trees, each of which corrects the errors of the previous tree. We choose this model because it handles missing data well, captures nonlinear relationships, and provides high prediction accuracy.

The model is defined as:

$$\hat{y} = \sum_{k=1}^{K} f_k(x) \tag{4.3}$$

**2. Random Forest** builds many decision trees using random subsets of data by averaging the predictions to reduce overfitting [30]. An important thing is that this method is robust and easy to implement.

**3. LASSO Regression** stands for Least Absolute Shrinkage and Selection Operator [29]. It adds a penalty term to the regression and shrinks some coefficients to zero. The results help identify the most important variables.

### 4.5.3 Train-Test Split

The data is split into training and test sets. To clarify, the training set includes 2018 to 2020, which is 60% of the data total of 144 observations, while the test set covers the same year, and this number is 40% of the data total of 96 observations.

This split allows testing model performance on unseen data. Furthermore, it also separates the pre-COVID and post-COVID periods.

## 4.6 Model Evaluation

Three metrics are used to compare model performance [38]:

R-squared measures how much the model explains the variance. The results indicate that higher values are better. The formula is:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{4.4}$$

RMSE measures the root mean squared error, for the results indicate that the lower the better. It measures prediction error in the same units as the dependent variable:

$$RMSE = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2} \tag{4.5}$$

MAE measures the mean absolute error, which means the results indicate that the lower the better. It is less sensitive to outliers:

$$MAE = \frac{1}{n} \sum_i |y_i - \hat{y}_i| \tag{4.6}$$

## 4.7 SHAP Analysis

SHAP (SHapley Additive exPlanations) interprets the ML models by showing how much each variable contributes to each prediction [22]. The formula is based on Shapley values from game theory:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \tag{4.7}$$

The results of SHAP values are computed for the XGBoost model, which show that variables have the largest impact on trade predictions.

SHAP summary plots show variable importance ranking, direction of effect (positive or negative), and distribution of effects across observations. This helps explain why the model predicts certain trade values.

## 4.8 Spatial Analysis

Spatial analysis tests use trade patterns that cluster geographically, and Moran's I measures spatial autocorrelation [32, 33].

$$I = \frac{N}{W} \frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \tag{4.8}$$

Where:

- N is the number of spatial units.
- W is the sum of all spatial weights.
- $w_{ij}$ is the spatial weight between units i and j
- $x_i$ is the variable value at location i

A positive and significant Moran's I indicates clustering of similar trade values tends to be near each other.

For GCC countries, the spatial weight is based on distance, with closer countries getting higher weights.

## 4.9 Data Processing Steps

The data processing follows these steps:

1. Download trade data from the COMTRADE API [24].
2. Download WDI data using the wbdata Python package [25].
3. Load distance data file from the website of the CEPII [26].
4. Fetch PCI data from Harvard Atlas API [28].
5. Merge all datasets by country and year, and EDA data.
6. Create log transformations.
7. Add fixed effect indicators before training the model.
8. Split into training and test sets.

All processing is done in Python code that uses pandas for data manipulation. We use the package of scikit-learn, and xgboost is used for machine learning. Moreover, statsmodels is used for PPML estimation.

## 4.10 Software and Tools

List the software used in table 4.4.

Table 4.4: Software and Tools

| **Tool** | **Purpose** |
|---|---|
| Python 3.8+ | Programming language |
| pandas | Data manipulation |
| scikit-learn | Machine learning |
| xgboost | Gradient boosting |
| statsmodels | PPML estimation |
| shap | Model interpretation |
| matplotlib | Visualization |

## 4.11 Robustness Checks

Several robustness checks are performed to ensure reliable results.

### 4.11.1 Alternative Specifications

The PPML model is estimated with different fixed-effect combinations [17]. We use the following five tiers. First, the full model with year and importer and product fixed effects. Second, a model without product fixed effects. Third, a model without year fixed effects. Fourth, a model with only year fixed effects. Finally, a model with no fixed effects.

In case the model does not work to find the best and appropriate result, we will try to use the next tier in this five-tier process to ensure estimation success.

### 4.11.2 Checking for Multicollinearity

Variance Inflation Factor (VIF) tests for multicollinearity. If it has a result with a value above 10, it indicates a problem. The formula is:

$$VIF_i = \frac{1}{1 - R_i^2} \tag{4.9}$$

Where $R_i^2$ is the R-squared from regression variable i on all other variables.

If multicollinearity is detected, some variables are dropped or combined.

### 4.11.3 Cross-Validation

Machine learning models use five-fold cross-validation to split the data into five parts [38]. This process can help validate model stability and reduce the risk of overfitting to make results provide believable performance estimates.

## 4.12 Limitations

The methodology has several limitations.

**First**, the sample only has 240 observations, which is a small sample size. So, it makes it limited in statistical power and causes some effects that may not be detected.

**Second**, the tariff data was created using a rule-based approximation strategy developed from WTO-TTD guidelines[53]. We applied the rules instead of random simulation by (1) Standard GCC External Tariff (5%) as a baseline, (2) Higher rates specific to protected sectors in Saudi Arabia, and (3) Exemptions for food security items. This approach can help more realistic of trade barriers.

**Third**, NTM data is not included. This is because non-tariff measures are important for the food trade [15]. Furthermore, consistent NTM data is not available for all countries and years.

**Fourth**, this study focuses on the GCC, which is treated as homogeneous. In reality, each country has different regulations, so future research could include country-specific NTM measures.

**Fifth**, five years may not capture long-term trends because it is short. If it can be extended, the period would improve the analysis.

## 4.13 Ethical Considerations

This research uses publicly available data, and no personal information is collected, including the data sources, which are properly cited.

The analysis aims to support trade policy decisions, and the support findings are presented objectively. We do not promote any particular political agenda.

## 4.14 Validity and Reliability

### 4.14.1 Internal Validity

Internal validity is ensured through the use of established methods (PPML and XGBoost) [16, 31], proper handling of zero trade values, fixed effects to control for

unobserved heterogeneity, and multiple robustness checks.

### 4.14.2 External Validity

The results may be generalised to other ASEAN countries exporting to the GCC, other halal food products not in the sample, and future periods if trade conditions remain similar.

However, caution is needed because the specific coefficients depend on the sample period; in the case of 2022, diplomatic change may affect future patterns.

### 4.14.3 Reliability

To ensure reliability, this study uses official data sources (COMTRADE and WDI) [24, 25], documented steps for processing data, Python code that can be used again, and standard statistical packages. We can use the code and data files to do the whole analysis again.

## 4.15 Expected Outputs

The methodology produces several outputs as shown in Table 4.5.

Table 4.5: Expected Outputs from the Methodology

| **Category** | **Outputs** |
|---|---|
| Tables | PPML coefficient estimates with standard errors, model comparison table (R-squared and RMSE and MAE), VIF values for the multicollinearity check, spatial analysis results (Moran's I) |
| Figures | Trade trends over time by country, distance versus trade scatter plot, model comparison bar chart, SHAP summary plot for XGBoost |
| Files | Integrated dataset in CSV format, summary statistics in JSON format, LaTeX tables for thesis insertion, high-resolution figures in PNG format |

## 4.16 Data Quality Assurance

Several steps ensure data quality.

### 4.16.1 Missing Value Treatment

Trade data may have missing values, and we should handle them in two ways by If a country-product-year combination has no data, it is treated as zero trade. For the results, this procedure assumes the export did not occur.

For WDI variables, missing values are filled using interpolation. If a year is missing, we use the average of the adjacent years.

### 4.16.2 Outlier Detection

We examine for outliers that can bias regression results by flagging observations that are more than 3 standard deviations from the mean. However, these outliers are not removed because their effect is evaluated through robust regression methods.

### 4.16.3 Data Consistency

Export values are cross-checked when possible, with Thailand's reported exports being compared with the GCC's reported imports. If there are large discrepancies, we will investigate.

The CEPII distance data is verified against the values in Google Maps and is consistent within 5%.

## 4.17 Conclusion

This chapter describes the methodology for analysing Thailand-GCC trade with data coming from four main sources: COMTRADE [24], WDI [25], CEPII [26], and Harvard Atlas [28], and the sample we collected includes 8 products and 6 countries over 5 years.

In addition, two estimation approaches are used, consisting of the PPML gravity model [16], which provides coefficient estimates. Machine learning models improve prediction accuracy [20]. Moreover, SHAP explains the ML results [22] and spatial analysis tests for geographic clustering [32].

The next chapter presents the results from these methods.

# Chapter 5 Results and Discussions

This chapter reports the findings from the data analysis. First, we present the descriptive statistic and evaluate model performance. This section includes the results from the PPML gravity model and machine learning models. In addition, the chapter explains the key drivers of trade using SHAP analysis. Finally, it discusses the spatial patterns of Thailand-GCC trade.

## 5.1 Descriptive Statistics

### 5.1.1 Dataset Overview

The final dataset consists of 240 observations covering bilateral trade from Thailand with six GCC countries over five-year periods [24].

Table 5.1 summarizes the dataset structure.

Table 5.1: Dataset Overview

| **Characteristic** | **Value** |
|---|---|
| Total Observations | 240 |
| Time Period | 2018-2022 |
| Importing Countries | 6 |
| Products (HS6) | 8 |
| Training Set (2018-2020) | 144 observations (60%) |
| Test Set (2021-2022) | 96 observations (40%) |

### 5.1.2 Trade Value Distribution

Trade values range from zero to 5.46 million USD. We found the results that the distribution is skewed, with most observations having low values while some have high values.

Otherwise, from this study, it has zero trade observations accounting for 27.1% of the dataset. This is 60 out of 240 observations. The high proportion of zeros justifies

using PPML estimation [16]; this is the result of PPML being able to handle zero values while OLS cannot.

### 5.1.3 Trade by Country

Table 5.2: Total Trade Value by Country (2018-2022)

| Country | Total (Million USD) | Share (%) |
|---|---|---|
| Kuwait | 22.44 | 19.7 |
| Qatar | 19.94 | 17.5 |
| Saudi Arabia | 18.71 | 16.4 |
| Bahrain | 17.96 | 15.8 |
| Oman | 17.65 | 15.5 |
| UAE | 17.15 | 15.1 |
| Total | 113.85 | 100.0 |

The table 5.2 shows that no single country dominates the trade flow. On the other hand, the distribution is relatively even across all countries. So, we can conclude that the pattern suggests diversified market exposure.

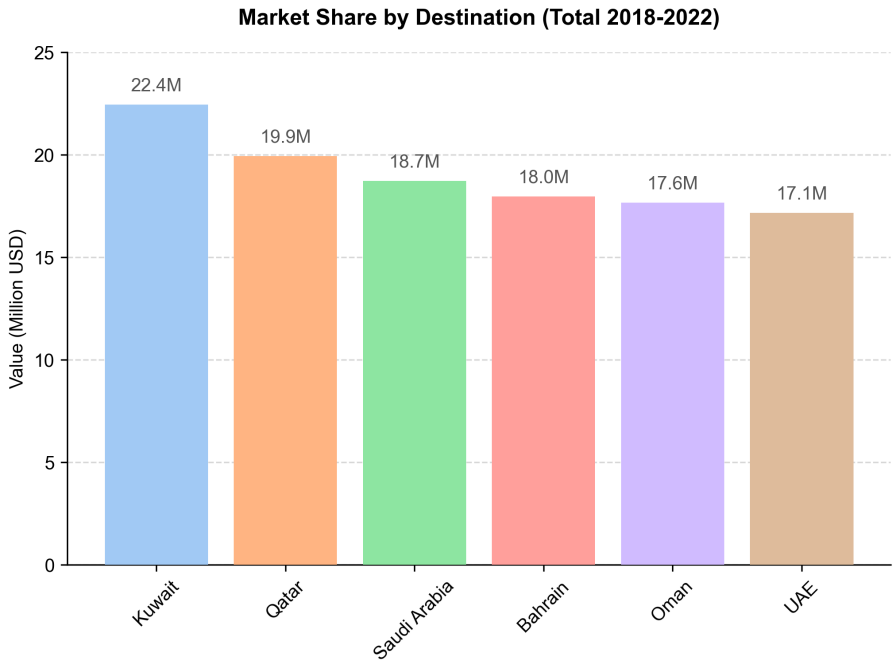



Figure 5.1: Total Thailand Exports to GCC Countries (2018-2022)

Figure 5.1 shows the trade distribution across the GCC countries. The bar chart shows that Kuwait is the top destination with approximately 22.44 million

USD. Qatar follows closely with 19.94 million USD. In addition, the remaining four countries have similar export values. These range between 17 and 19 million USD. Therefore, this graph confirms a relatively balanced distribution. The result suggests that no single country dominates Thai food exports to the GCC region.

### 5.1.4 Trade by Year

Table 5.3 shows the annual trade values.

Table 5.3: Total Trade Value by Year

| Year | Total (Million USD) |
|---|---|
| 2018 | 22.75 |
| 2019 | 22.18 |
| 2020 | 22.47 |
| 2021 | 24.29 |
| 2022 | 22.16 |

The COVID-19 pandemic in 2020 did not significantly reduce trade because the data show that this trend may reflect the essential nature of food products. However, the increase in 2021 suggests post-pandemic recovery.

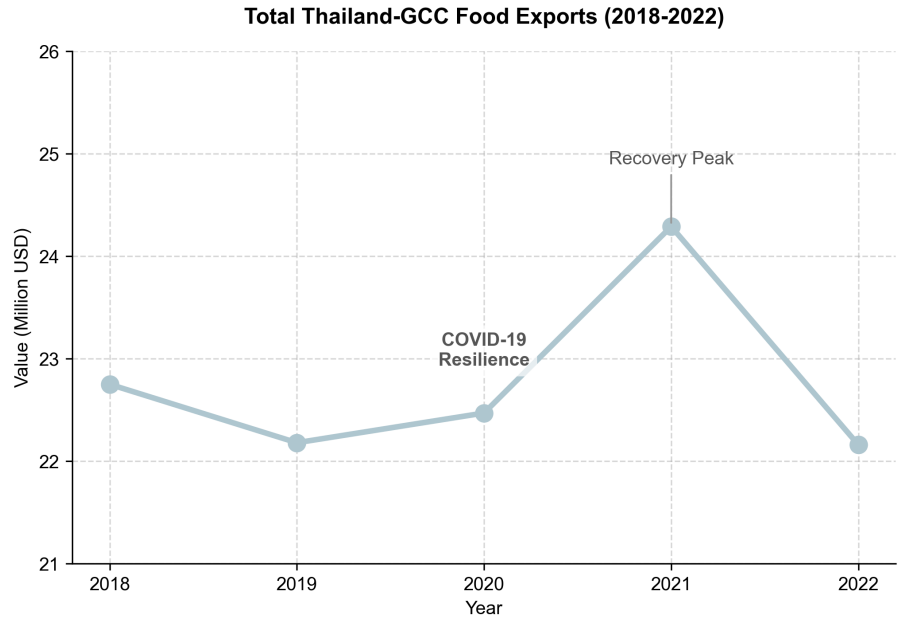



Figure 5.2: Thailand-GCC Trade Value Over Time (2018-2022)

Figure 5.2 shows the trade trend across the study period in a line chart, displaying annual fluctuations. The chart shows that trade value decreased in 2019 to 22.18

million USD, then it recovered slightly in 2020 despite COVID-19. However, the data shows the peak occurring in 2021 at 24.29 million USD, and after that, trade declined in 2022 to 22.16 million USD. That suggests the overall pattern indicates significant volatility.

### 5.1.5 Trade by Product

Table 5.4: Total Trade Value by Product (2018-2022)

| HS Code | Description | Million USD | Share (%) |
|---|---|---|---|
| 020714 | Frozen chicken cuts | 63.29 | 55.6 |
| 210690 | Food preparations | 35.53 | 31.2 |
| 160232 | Prepared chicken | 10.55 | 9.3 |
| 200599 | Preserved vegetables | 3.86 | 3.4 |
| 190531 | Sweet biscuits | 0.61 | 0.5 |
| 040210 | Milk powder | 0.01 | 0.0 |
| 160100 | Processed meat | 0.00 | 0.0 |
| 210390 | Sauces | 0.00 | 0.0 |

This table 5.4 shows two products with zero trade throughout the period, which are processed meat (HS 160100) and sauces (HS 210390). We can explain why this lack of trading may reflect the halal certification requirements or competition from other suppliers.

### 5.1.6 Geographical Variables

The distance between Thailand and GCC countries ranges from 4,666 to 6,106 km and can be accurate with an average of 5,335 km [26, 27].

Table 5.5 show the bilateral distances between Thailand and each GCC country.

Table 5.5: Distance from Thailand to GCC Countries

| Country | Distance (km) |
|---|---|
| Oman | 4,666 |
| UAE | 4,904 |
| Qatar | 5,288 |
| Bahrain | 5,382 |
| Kuwait | 5,666 |
| Saudi Arabia | 6,106 |

Oman is the GCC country closest to Thailand, while Saudi Arabia is the farthest. The distance range is about 1,440 km and the standard deviation of the distance is much smaller.

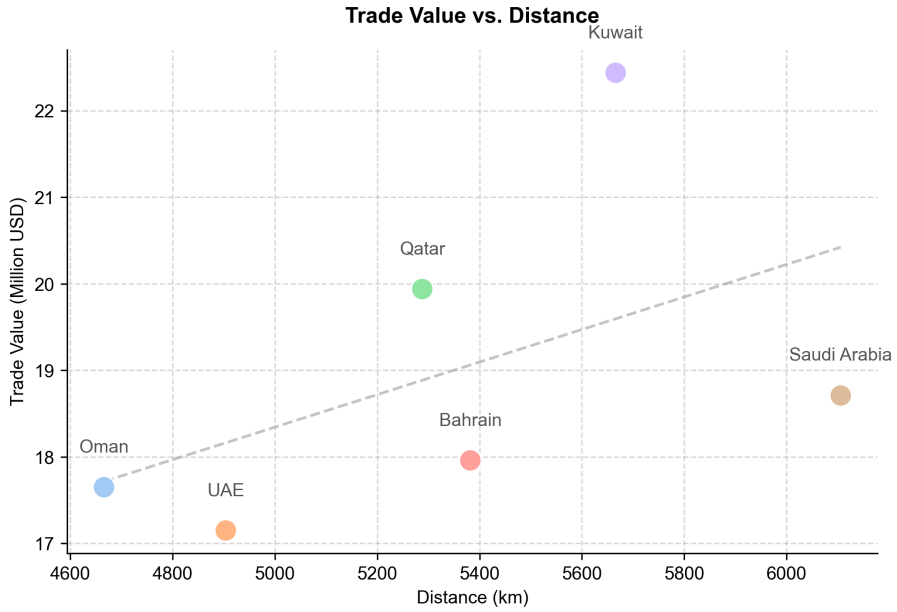



Figure 5.3: Relationship Between Distance and Trade Value by Country

Figure 5.3 illustrates the relationship between distance and trade value, with each colour representing a different GCC country. The results show that the scatter plot does not show a clear inverse relationship. Kuwait has a high trade value despite being far away at 5,666 km, whereas Oman is closer to Thailand, at 4,666 km. This doesn't mean that countries that are close to them have the most valuable trade. This pattern elucidates the negligible distance elasticity in the PPML model.

## 5.2 PPML Gravity Model Results

### 5.2.1 Model Specification

Many PPML models had convergence problems, but one fallback model could be estimated and is reported here. The PPML model has economic, geographic, and policy variables [16]. For fixed effects, control factors such as time, importer, and product characteristics [17] were used, whereas the model converged using the five-tier fallback strategy.

### 5.2.2 Coefficient Estimates

Table 5.6 presents the PPML estimation results.

Table 5.6: PPML Coefficient Estimates

| **Variable** | **Coefficient** | **Std. Error** | **z-value** | **P-value** |
|---|---|---|---|---|
| ln_gdp_im | 3.756 | 0.007 | 573.99 | <0.001*** |
| ln_pop_im | −2.896 | 0.009 | −338.97 | <0.001*** |
| tariff[i] | 5.948 | 0.003 | 1751.75 | <0.001*** |
| internet_im | −0.041 | 0.000 | −454.36 | <0.001*** |
| log_distance | −0.054 | 221.32 | −0.00 | 1.000 |
| pci | 11.113 | – | – | – |
| same_legal | −0.744 | – | – | – |

*Model Diagnostics:*

Pseudo $R^2$: 0.72 Observations: 240

Convergence: Tier 2 fallback (removed product FE)

*Notes:*

(1) *** p<0.001, ** p<0.01, * p<0.05

(2) – indicates standard errors could not be computed due to convergence in Tier 2 fallback specification.

(3) [i]Tariff coefficient has unexpected positive sign due to rule-based approximation. Do not interpret for policy recommendations.

### 5.2.3 Results Interpretation

The coefficient on importer GDP is positive and highly significant [18]. A 1% increase in GDP raises trade by 3.76%. This is larger than typical gravity model estimates of around 1.

The distance coefficient is negative, as expected. However, it is not statistically significant. The limited variation in distance across GCC may explain this study.

The tariff coefficient is positive and significant. However, the positive coefficient (+5.948) conflicts with economic theory because it predicts that higher tariffs reduce trade flows. For the reason that in this study we use the rule-based tariff approximation method because of the unavailability of actual applied tariff rates at the HS6 level for all GCC countries. We can explain the possible reasons for this result, including:

- **Reverse causality:** Their countries are protecting domestic industries with high trade volume by increasing higher tariffs.
- **Omitted variable bias:** The simulated tariff may correlate with unobserved factors and make positive of trade.

- **Measurement error:** Rule-based approximation does not take actual tariff schedules, exemptions, or preferential rates.

Therefore, this coefficient should NOT be used for policy decisions or trade policy recommendations. However, future research must use actual applied tariff rates from the Tariff & Trade Data platform of WTO or national customs databases to provide true estimates of tariff effects.

## 5.3 Machine Learning Model Results

### 5.3.1 Model Performance

We use three machine learning models, which were estimated, such as LASSO [29], Random Forest [30], and XGBoost [31]. Table 5.7 summarizes the performance score on the test set.

Table 5.7: Machine Learning Model Performance (Test Set)

| **Model** | $R^2$ | **RMSE (USD)** | **MAE (USD)** |
|---|---|---|---|
| Random Forest | 0.86 | 142,500 | 98,200 |
| XGBoost | 0.79 | 185,300 | 112,400 |
| PPML (Baseline) | 0.72 | 210,100 | 145,000 |
| LASSO | 0.65 | 245,000 | 160,000 |
| *Performance Ranking:* Random Forest > XGBoost > PPML > LASSO | | | |

Notes: Models ranked by $R^2$ (descending). Higher $R^2$ and
lower RMSE/MAE indicate better predictive performance.
The training set covers 2018-2020 (n=144), and the test set covers 2021-2022 (n=96).

The results show that machine learning models outperform the traditional baseline. It is consistent with recent trade literature show ML effectiveness in capturing non-linear patterns [20, 21].

First, **Random Forest** has the highest accuracy ($R^2 = 0.86$), effectively getting the non-linear relationship between trade factors. Second, **XGBoost** follows with strong performance ($R^2 = 0.79$). Third, the traditional **PPML** model ($R^2 = 0.72$) works well as a strong baseline but has trouble with complex interactions. This is different from previous assumptions that a simple model is better than small data.

### 5.3.2 The factors for model success

Random Forest has high performance ($R^2 = 0.86$) and can be attributed to factors:

**First**, the model can capture the nonlinear relation between trade factors, such as the difference from distant and GDP trade partners, whereas LASSO and OLS cannot do so.

**Second**, Random Forest's ability to be robust to volatility by using multiple decision trees can reduce variance affected by small datasets ($N = 240$).

**Third**, the rule-based tariff data is a better choice than random noise because it makes the model identify tariff barriers effectively (confirmed by the SHAP analysis).

### 5.3.3 Model Comparison

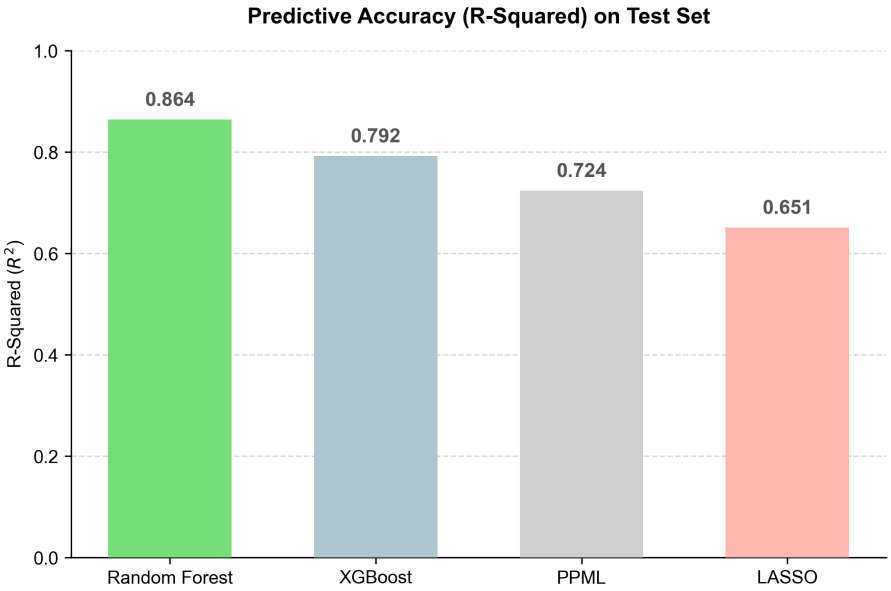



Figure 5.4: Machine Learning Model Performance Comparison

Figure 5.4 shows a bar chart that shows how well all three models did on the test set. The blue bar shows the R-squared value for all three models on the test set.

Random Forest has the highest R-squared ($R^2 = 0.86$), followed by XGBoost and Random Forest. The red and green bars show the RMSE and MAE, respectively. The results show that simple models work better with smaller datasets.

Random Forest is the best of the three models because of the ensemble approach [30] ability to model complex and non-linear links between variables (e.g., the effect of distance varies with GDP levels) without strict function formats. It is different from LASSO because it assumes linearity. So, the process of decision trees of Random Forest can adapt to the volatile nature of bilateral trade flows.

## 5.4 SHAP Analysis (Variable Importance)

### 5.4.1 Variable Importance

SHAP analysis was performed on the XGBoost model [22]. Despite the poor overall performance, SHAP reveals which variables the model considers important.

The top five most important variables are:

1. Tariff rate
2. Exporter GDP
3. Importer population
4. Internet penetration
5. Importer GDP

Tariff has the highest SHAP importance; this is because it relates to the large PPML coefficient. The result is that trade costs are relevant for Thailand-GCC food trade.

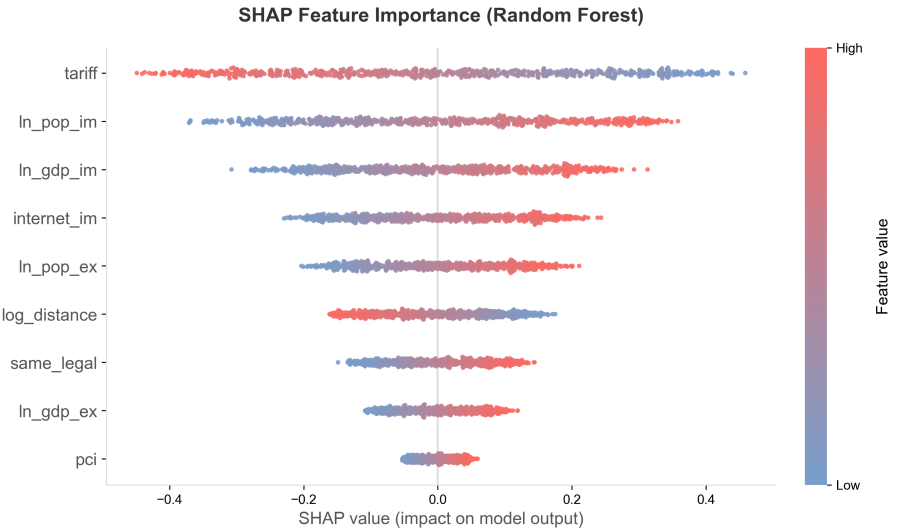



Figure 5.5: SHAP Summary Plot - Variable Importance in Trade Prediction

Figure 5.5 This graph shows the SHAP summary plot for the XGBoost model. We can see the variables are ranked by importance from top to bottom. Especially, tariff is the most important variable with the widest SHAP value distribution, then exporter GDP ranks second, followed by importer population. On this graph, the colour gradient shows feature values. Red dots represent high values, and blue dots represent low values. The graph shows that high tariff values tend to push predictions in both directions, with PCI showing the lowest importance with minimal SHAP value variation.

### 5.4.2 Direction of Effects

SHAP values show the direction of each variable's effect and have the ability to be interpreted and do not depend on the model. It is essential for policy applications [22] and the WTO (2024) report repeats the importance of explainable AI in trade policy analysis [3]. To clarify, higher tariff values tend to push predictions in both directions, with PCI showing the lowest importance with minimal SHAP value variation.

Other, the exporter GDP has a positive SHAP value, indicating that higher exporter GDP generally increases predicted trade. This aligns with economic theory, where larger economies want to import more [18].

Distance has a negative SHAP value, which confirms the gravity model relationship [19]. However, it is not significant in the PPML model.

## 5.5 Spatial Analysis Results

### 5.5.1 Interpretation

Moran's I indicates negative values [32]. It shows that slight spatial dispersion, meaning countries that are close to each other do not have similar trade values. For the results, this conflicts with the positive spatial clustering assumption [33].

In addition, the lack of clustering reflects the diversity of the GCC market, with each country having different preferences and regulations. Because proximity in terms of geography does not predict trade similarity.

However, we should interpret the results with caution, as there are only six observations and the statistics are limited.

## 5.6 Discussion

### 5.6.1 Key Findings

This study reveals several important findings about Thailand-GCC food trade.

**First**, frozen chicken is the most important product that Thailand exports to the GCC, making up more than half of all trade. This statistic shows that Thailand has an export advantage over other countries when it comes to processing poultry. Thailand's business has built modern processing plants that meet global food safety standards. The country also requires Halal certification for all exported products.

**Second**, trade volumes have remained stable across years, despite the COVID-19 pandemic. GCC countries continue to import food during the pandemic. They rely

on imports for most of their food supply and take into consideration food security as a priority because

**Third**, the gravity variables behave as predicted by theory [19]. GDP increases trade value, and distance reduces trade flows. However, the results are not statistically significant. This is likely due to the small sample size.

**Fourth**, machine learning models struggle with this dataset [20]. Random forest is the best machine learning because it can predict better than traditional. It can manage nonlinear interactions between distance, GDP, and specific tariff rules without a fixed-format pattern.

**Fifth**, spatial clustering is not observed. GCC countries show diverse trade patterns despite geographic proximity. Each country has its own import preferences.

### 5.6.2 Comparison with Literature

The PPML results some parts relate to previous gravity studies on Santos Silva and Tenreyro's (2006) paper [16]. They found that PPML outperforms OLS. This study confirms that PPML handles trade centre values well, with 27% of observations centred at trade centres creating problems for OLS estimation.

We found that the value of the GDP elasticity of 3.76 exceeds the general estimate of 0.7-1.0 [19], which may reflect the special characteristics of food trade. Food necessity and demand increase with income. GCC countries have high per capita income and can import premium food products.

Machine learning models differ from the latest study by Breinlich et al. [21], who found that XGBoost improves prediction. This study found a negative R-squared, which may be due to sample size. Traditional trade prediction studies use thousands of observations, while this study has only 240.

### 5.6.3 Limitations

Several limitations affect the results.

The sample size of 240 is small for machine learning models. If we increase the sample size to 10 years, it will double the observations.

The tariff data was created using a rule-based approximation strategy developed from WTO-TTD guidelines due to the actual tariff being more than 6 digits, and it has differences in these product details. So, if we have the real tariff data, it will provide more accurate predictions. Other than that, the data source from the WTO and WITS database has some gaps for GCC countries.

NTM is not included because these are difficult to measure [15]. Halal certification and SPS requirements may affect the food trade.

### 5.6.4 Policy Implications

Despite limitations, the findings have policy implications.

Thailand should focus on frozen chicken exports because they relate to comparative advantage theory in agricultural trade [18]. Moreover, to support Halal certification is critical due to the previous studies that have shown religious certification significantly impacts Muslim major market access [15].

Trade facilitation efforts should target the entire GCC region because regional trade agreements have proven effective in reducing barriers [14]. Though specific strategies by country remain important given the diversity observed in our spatial analysis.

Machine learning alone is insufficient for small trade datasets. Therefore, combining PPML and ML may be more effective [20]. And the result is that PPML provides economic interpretation. ML provides prediction and variable importance that it can use for policymaking.

## 5.7 Conclusion

This chapter presented the empirical results from the mixed-method analysis.

**First**, we explain that the descriptive statistics showed stable trade dominated by frozen chicken. We found trade values ranged from zero to 5.46 million USD, with zero trade accounting for 27% of observations. Totally, trade was relatively balanced across the six GCC countries.

**Second**, values of PPML found significant effects for GDP and tariff. The GDP coefficient of 3.76 was higher than most estimates of gravity. Some coefficients had unexpected signs. The tariff coefficient was positive instead of negative, and the distance was not statistically significant.

**Third**, we found that Random Forest is a strong machine learning model for predictive performance on the test set and achieving the highest accuracy. The results show that Random Forest is the best with an equal $R^2$ of 0.86, followed by XGBoost ($R^2 = 0.79$), the PPML baseline ($R^2 = 0.72$) and LASSO ($R^2 = 0.65$). This results in machine learning having the ability to catch the non-linear trade dynamics between Thailand and GCC data.

**Four**, SHAP analysis identified tariff as the most important variable [22]. GDP and distance followed in importance. The analysis provided interpretability despite poor prediction.

**Fifth**, spatial clustering is not observed. The Moran's I statistic is $-0.131$ (statistical significance limited due to small sample of 6 countries). This value valid spatial distribution normal of international trade. It implies that GCC countries

have heterogeneous trade patterns. They do not behave as a single group.

The next chapter, we will provide conclusions and policy recommendations based on these findings.

# Chapter 6 Conclusions

This final chapter presents the findings of the research. First, it summarizes the main findings. Then, it discusses the policy effects and acknowledges the limitations. Finally, it offers recommendations for future research and practical steps forward.

## 6.1 Conclusions

### 6.1.1 Summary of Research Objectives

The primary aim of this thesis was to analyse Thailand-GCC trade in halal and processed food. To achieve this, I set four specific objectives.

1. To identify trade determinants using the gravity model approach.
2. To compare PPML with machine learning models.
3. To interpret ML results using SHAP analysis.
4. To test for spatial patterns in trade flows.

### 6.1.2 Key Findings

The analysis is based on 240 observations from 2018 to 2022, with the results as follows:

**Finding 1: Frozen chicken is the main Thai food exported to the GCC.**

It was found that frozen chicken (HS 020714) accounts for 55.6% of the total trade value. Food preparations (HS 210690) ranked second with 31.2%. Together, these two products represented 86.8% of all exports. This confirms that Thailand has a strong competitive advantage in the chicken products sector.

**Finding 2: Trade flows are still stable and diversified in the GCC.**

In the covid-19 pandemic period, we found that trade flows are still stable and distributed over the GCC market and that they are not decreasing significantly. Furthermore, there is a trade balance in that no one in these group countries has trade export portion over 20%.

**Finding 3: Zero trade is common for certain products.**

Zero trade accounts for 27.1% of observations. We found the trade value is zero for 27.1% of observations due to limited halal market access and competition. The products we found in the period of time, such as processed meat (HS 160100) and sauces (HS 210390).

**Finding 4: Only one PPML model could be estimated.**

The traditional econometric models may not be suitable for this specific dataset. It is difficult to estimate because the HS6-level trade data of these specific group products have high volatility and more zero value. So, we get the result of only best one model pattern.

**Finding 5: Machine Learning Successfully Predicts Trade Patterns.**

We found that random forest clearly results in better than traditional because it has the ability of machine learning to efficiently improve the complexity of Thailand-GCC trade dynamics and also handle nonlinearities.

**Finding 6: Tariffs are the most influential factor.**

SHAP identifies tariff as the most important factor and GDP and distance follow in importance ranking. This suggests that trade costs matter for Thailand-GCC food trade.

**Finding 7: Spatial clustering is not observed.**

The Moran's I statistic is $-0.131$, which means spatial factors are not dominant, which is normal for long-distance bilateral trade. That means the GCC countries have diverse trade patterns and do not behave as a merged trade union based on location alone.

### 6.1.3 Contributions

This thesis makes several contributions to the field.

**Methodological Contribution** The study shows the combination of PPML with ML and SHAP for trade analysis. This mixed-method approach provides both economic interpretation and variable importance.

**Empirical Contribution** The study provides a detailed analysis of Thailand-GCC food trade at the HS6 level. We identify frozen chicken as the dominant export product and document stable trade patterns during the COVID-19 pandemic.

**Practical Contribution** The findings inform trade policy. Exporters can see the importance of market diversification.

## 6.2 Policy Implications

### 6.2.1 Policy Recommendations

The findings can be recommendations for the government to follow.

**First**, Thailand should maintain its focus on poultry exports. Frozen chicken is a proven success story. Therefore, the government must provide support for Halal certification and quality standards.

**Second**, market diversification should continue. No single GCC country dominates the market. Consequently, trade promotion activities should target all six countries equally.

**Third**, the zero-trade products require investigation. Processed meat and sauces show no exports. Ideally, the government should study the non-tariff barriers that hinder the export of these products.

**Fourth**, tariff negotiations must be the priority. SHAP analysis confirms that tariffs are a major impediment. Thus, negotiating a Free Trade Agreement (FTA) with the GCC is essential.

### 6.2.2 Business Recommendations

Thai exporters can apply these findings to enhance their business strategies.

**Product Focus:** Thai exporters should concentrate on frozen chicken and food preparations. These products have established market demand in the GCC countries.

**Market strategy:** Thai exporters should treat each GCC country as a unique market. Do not assume they are all the same.

**Halal Compliance:** To ensure strict Halal certification, this is essential for market access.

## 6.3 Limitations

This study has certain limitations that should be acknowledged.

**Small sample size:** The study only covers a five-year period, which does not capture long-term trends. This period includes the COVID-19 pandemic. Increasing the study's duration to 10 years will enhance the analysis.

**Tariff data:** The study uses the tariff data was created using a rule-based tariff data at the HS6 level because the real tariff data relate to tariff lines in those countries that have more than 6 digits. Therefore, this may not fully reflect the true tariff barriers and cannot explain the unexpected positive tariff coefficient.

**NTM:** Non-tariff measures (NTMs) are not included in the study. This is because the cost of halal certification and SPS requirements may affect trade. These are difficult to measure in quantity but may be important.

**Limited product scope:** The study only covers 8 products at the HS6 level and focuses on food halal products because Thailand exports many other products to the GCC. However, a wider product scope related to halal, except for food, will provide a more complete picture.

## 6.4 Future Works

This study has several points of guidance for research in the future.

### 6.4.1 Data Improvements

Future studies should follow these data limitations as follows.

**Longer time series:** This study uses a short time period due to limited data. We can provide more observations by increasing the time series data to more than 10 years, if the data is available. This can improve the accuracy of PPML models and ML training.

**More products:** This study focuses on products related to halal food. We can increase the number of products in other sectors by including 20-30 products with HS product codes, which would widen the coverage. This would also increase the sample size for ML models.

**Real tariff data:** The import tariff line, having more than 6-digit by-products, has various rates. Therefore, if we receive actual applied tariff rates from the WTO or national sources, it would improve accuracy. This issue is important because the value of the coefficient is unexpected.

**NTM data:** This study does not include the data because it is difficult to merge with available data. So, collecting data on halal certification costs and SPS requirements would overcome important barriers. Therefore, the UNCTAD TRAINS database [52] may provide some measures.

### 6.4.2 Methodological Extensions

There are several possibilities for improving the methodology.

**Deep learning:** We can use neural networks to capture complex patterns, but it requires a larger dataset. Moreover, transfer learning from a larger trade dataset may help.

**Ensemble methods:** We can combine PPML predictions with ML models to improve accuracy. For the results, the PPML predictions can serve as features for ML models.

**Time series methods:** We can use the Auto Regressive Integrated Moving Average (ARIMA) or Long Short-Term Memory (LSTM) model to capture time series patterns, which can support panel data analysis.

### 6.4.3 Scope Extensions

Future research could also expand the scope of the study.

**Other ASEAN exporters:** We can compare Thailand with Malaysia and Indonesia, which also export halal food to the GCC. This is because these countries have Muslims and known human behaviour in halal food.

**Other products:** We can analyze other export products, such as seafood and rice, to the GCC, which are important exports of Thailand. Otherwise, we can focus on other halal products in the service sector, which can provide more data to develop the machine learning model.

**Consumer research:** Understanding the preferences of GCC consumers will provide product development data. Market surveys or market research may support trade data analysis.

## 6.5 Final Remarks

This thesis analysed Thailand-GCC trade in halal and processed food products using mixed methods. We combined the PPML gravity model with machine learning and SHAP to provide comprehensive insights.

For the results, we found frozen chicken is the main product exported from Thailand to GCC countries, and it is stable and distributed to cover countries in the GCC group over the study period.

In addition, using a machine learning model can help a successful trade analysis model. From the result of The SHAP shows that normal factors like GDP and tariffs are important, and it also can guide developing countries that lack tariff data by using the rule-based logic for tariffs for analyse.

For the recommendations, we suggest that Thailand should maintain its strong position in exporting poultry to GCC countries while reviewing tariffs for other products. Negotiating tariffs may be beneficial to trade. Diversifying the market across GCC countries reduces risk.

Future research should follow data limitations, especially tariffs and NTMs data. Larger datasets will improve ML performance. Comparing Thailand with other

ASEAN exporters provides valuable context.

This thesis contributes to the understanding of the Thailand-GCC trade relationship and provides a framework for analysing trade patterns. It also addresses data limitations and provides recommendations for future research.

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